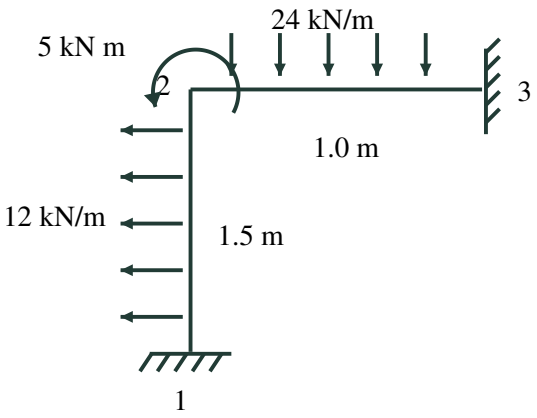
Question 1 · Rigid-jointed frame

Frame geometry and restraints

Analyse the rigid-jointed frame using plane-frame elements. Determine the two translations and the rotation of the corner joint. All members have the same section and elastic properties.



The column runs from fixed node 1 to joint 2. The beam runs from joint 2 to fixed node 3. Take right, upward and counterclockwise as positive. Read the distributed loads as leftward on the column and downward on the beam.

$$l_1 = 1.5\,\mathrm{m}, \quad l_2 = 1\,\mathrm{m}, \quad q_1 = 12\,\mathrm{kN/m}, \quad q_2 = 24\,\mathrm{kN/m}.$$

$$M_2 = +5\,\mathrm{kN\,m}, \qquad \mathbf{d}_f = \begin{bmatrix} u_2 & v_2 & \theta_2 \end{bmatrix}^T.$$

Question 1 · Rigid-jointed frame

Local member stiffness

For a prismatic Euler–Bernoulli frame member of length L, take the local x-axis along the member from its first node i to its second node j. Take local y at 90° counterclockwise to local x.

$$\mathbf{d}' = \begin{bmatrix} u'_i & v'_i & \theta_i & u'_j & v'_j & \theta_j \end{bmatrix}^T.$$

$$a = \frac{EA}{L}, \quad b = \frac{12EI}{L^3}, \quad c = \frac{6EI}{L^2}, \quad d = \frac{4EI}{L}, \quad h = \frac{2EI}{L}.$$

$$\mathbf{k}' = \begin{bmatrix} a & 0 & 0 & -a & 0 & 0 \\ 0 & b & c & 0 & -b & c \\ 0 & c & d & 0 & -c & h \\ -a & 0 & 0 & a & 0 & 0 \\ 0 & -b & -c & 0 & b & -c \\ 0 & c & h & 0 & -c & d \end{bmatrix}.$$

Here EA is the axial rigidity and EI is the bending rigidity. Member end forces and local degrees of freedom follow the same order.

Question 1 · Rigid-jointed frame

Rotate into global axes

Let α be the counterclockwise angle from global x to the member axis. The matrix R converts one node's global translations and rotation into local components.

$$\mathbf{R} = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} \end{bmatrix}.$$

$$\mathbf{d}' = \mathbf{T}\mathbf{d}, \qquad \mathbf{k} = \mathbf{T}^T\mathbf{k}'\mathbf{T}, \qquad \mathbf{f} = \mathbf{T}^T\mathbf{f}'.$$

For the column, $\alpha = 90°$. Local axial displacement equals global vertical displacement and local transverse displacement equals negative global horizontal displacement.

$$\mathbf{R}_1 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For the horizontal beam, $\alpha = 0°$ and its local axes coincide with the global axes.

Question 1 · Rigid-jointed frame

Retain the free-end stiffness blocks

At both supports the prescribed displacements vanish. For the column, keep the second-node block after rotation. For the beam, keep the first-node block. The order in both blocks is horizontal, vertical, rotation.

$$\mathbf{K}_{1f} = \begin{bmatrix} 12EI/l_1^3 & 0 & 6EI/l_1^2 \\ 0 & EA/l_1 & 0 \\ 6EI/l_1^2 & 0 & 4EI/l_1 \end{bmatrix}.$$

$$\mathbf{K}_{2f} = \begin{bmatrix} EA/l_2 & 0 & 0 \\ 0 & 12EI/l_2^3 & 6EI/l_2^2 \\ 0 & 6EI/l_2^2 & 4EI/l_2 \end{bmatrix}.$$

The column resists horizontal motion by bending and vertical motion axially. The beam supplies axial stiffness horizontally and bending stiffness vertically.

Question 1 · Rigid-jointed frame

Assemble the corner stiffness

The same free joint belongs to both members. Add the corresponding entries of their stiffness blocks.

$$\mathbf{K}_f = \mathbf{K}_{1f} + \mathbf{K}_{2f}.$$

$$\frac{12}{(1.5)^3} = \frac{32}{9}, \quad \frac{6}{(1.5)^2} = \frac{8}{3}, \quad \frac{1}{1.5} = \frac{2}{3}.$$

$$\frac{4}{1.5} + \frac{4}{1} = \frac{20}{3}.$$

$$\mathbf{K}_f = E \begin{bmatrix} A + \frac{32}{9} I & 0 & \frac{8}{3} I \\ 0 & \frac{2}{3} A + 12 I & 6 I \\ \frac{8}{3} I & 6 I & \frac{20}{3} I \end{bmatrix}.$$

Lengths have been substituted in metres. Use A in square metres, I in metres to the fourth power and E in kilonewtons per square metre in all subsequent numerical-coefficient expressions.

Question 1 · Rigid-jointed frame

Convert distributed loads to joint loads

For a constant signed transverse load w in local y, the consistent member load vector carries a shear and a moment at each end.

$$\mathbf{f}' = \begin{bmatrix} 0 & wL/2 & wL^2/12 & 0 & wL/2 & -wL^2/12 \end{bmatrix}^T .$$

The column load acts in positive local y. At joint 2, the end shear points left and the end moment is clockwise.

$$\mathbf{f}_{1f} = \begin{bmatrix} -12(1.5)/2 \\ 0 \\ -12(1.5)^2/12 \end{bmatrix} = \begin{bmatrix} -9 \\ 0 \\ -9/4 \end{bmatrix} .$$

The beam load acts in negative local y. Joint 2 is its first end.

$$\mathbf{f}_{2f} = \begin{bmatrix} 0 \\ -24(1)/2 \\ -24(1)^2/12 \end{bmatrix} = \begin{bmatrix} 0 \\ -12 \\ -2 \end{bmatrix} .$$

Question 1 · Rigid-jointed frame

Form the three equilibrium equations

Add the applied corner moment to the two consistent load contributions. The first two entries are forces in kN and the third is a moment in kN m.

$$\mathbf{f}_f = \begin{bmatrix} -9 \\ 0 \\ -9/4 \end{bmatrix} + \begin{bmatrix} 0 \\ -12 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -9 \\ -12 \\ 3/4 \end{bmatrix}.$$

Write the five independent entries of the reduced matrix as follows.

$$p = E(A + 32I/9), \quad r = E(2A/3 + 12I), \quad t = 20EI/3,$$

$$c = 8EI/3, \qquad s = 6EI.$$

$$pu_2 + c\theta_2 = -9, \qquad rv_2 + s\theta_2 = -12,$$

$$cu_2 + sv_2 + t\theta_2 = 3/4.$$

Question 1 · Rigid-jointed frame

Eliminate the two translations

From the first and second equations, express the translations in terms of the single unknown rotation.

$$u_2 = \frac{-9 - c\theta_2}{p}, \qquad v_2 = \frac{-12 - s\theta_2}{r}.$$

Substitute these into the moment equation and gather the rotation terms.

$$\left(t - \frac{c^2}{p} - \frac{s^2}{r}\right)\theta_2 = \frac{3}{4} + \frac{9c}{p} + \frac{12s}{r}.$$

$$D = prt - rc^2 - ps^2, \qquad \theta_2 = \frac{3pr/4 + 9cr + 12sp}{D}.$$

$$H = 18A^2 + 223AI + 288I^2, \qquad D = \frac{20}{81}E^3IH.$$

For positive elastic and section properties, D is positive. Back-substitute this rotation into the two translation formulas.

Question 1 · Rigid-jointed frame

Unknown displacements and verification

With H as defined on the preceding page, the free corner displacements are

$$u_2 = -\frac{27(31A + 459I)}{5EH} \text{ m},$$

$$v_2 = -\frac{9(1521A + 6464I)}{40EH} \text{ m},$$

$$\theta_2 = \frac{9(9A^2 + 1778AI + 10368I^2)}{40EIH} \text{ rad}.$$

The corner moves left and downward and rotates counterclockwise. The two supported nodes have zero translations and rotations. Numerical displacement magnitudes require numerical values of E, A and I.

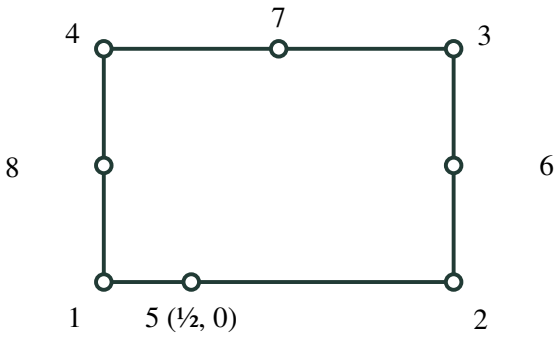
Substitution check using the exact expressions gives

$$\begin{bmatrix} pu_2 + c\theta_2 \\ rv_2 + s\theta_2 \\ cu_2 + sv_2 + t\theta_2 \end{bmatrix} = \begin{bmatrix} -9 \\ -12 \\ 3/4 \end{bmatrix}.$$

Question 2 · Serendipity rectangle

Eight-node rectangle

Determine the four corner shape functions by the serendipity construction. The eight specified positions are in physical coordinates.



$$(x_1, y_1) = (0, 0), \quad (x_2, y_2) = (2, 0),$$
$$(x_3, y_3) = (2, 2), \quad (x_4, y_4) = (0, 2).$$

$$(x_5, y_5) = (1/2, 0), \quad (x_6, y_6) = (2, 1),$$
$$(x_7, y_7) = (1, 2), \quad (x_8, y_8) = (0, 1).$$

For each corner i, construct a product of three straight-line factors that vanish collectively at the other seven nodes. Divide the product by its value at node i.

Question 2 · Serendipity rectangle

Construct the bottom-left function

For node 1, the line x = 2 covers nodes 2, 6 and 3. The line y = 2 covers nodes 4, 7 and 3. The remaining nodes 5 and 8 lie on the line 2x + y = 1.

$$g_1 = (x-2)(y-2)(2x+y-1).$$

$$g_1(0,0) = (-2)(-2)(-1) = -4.$$

$$N_1 = \frac{g_1(x,y)}{g_1(0,0)} = \frac{(2-x)(2-y)(1-2x-y)}{4}.$$

At node 5, the last factor is zero. At node 1, all three normalised factors give a product of one.

Question 2 · Serendipity rectangle

Construct the bottom-right function

For node 2, use x = 0 and y = 2 for the two opposite edges. The remaining two nodes are 5 and 6. Their joining line goes through (1/2, 0) and (2, 1).

$$y = \frac{1-0}{2-1/2}\left(x - \frac{1}{2}\right) = \frac{2}{3}x - \frac{1}{3}.$$

$$g_2 = x(y-2)(2x-3y-1), \qquad g_2(2,0) = -12.$$

$$N_2 = \frac{x(2-y)(2x-3y-1)}{12}.$$

The normalising denominator differs from the one for node 1 because the bottom-edge node is off-centre.

Question 2 · Serendipity rectangle

Construct the two top functions

For node 3, use x = 0, y = 0 and the line through nodes 6 and 7. For node 4, use x = 2, y = 0 and the line through nodes 7 and 8.

$$g_3 = xy(x + y - 3), \qquad g_3(2, 2) = 4.$$

$$N_3 = \frac{xy(x + y - 3)}{4}.$$

$$g_4 = (x - 2)y(y - x - 1), \qquad g_4(0, 2) = -4.$$

$$N_4 = \frac{(2 - x)y(y - x - 1)}{4}.$$

Each function vanishes along the bottom edge, which contains nodes 1, 5 and 2.

Question 2 · Serendipity rectangle

Check the eight nodal positions

Evaluate the four corner functions at the eight specified nodes. Rows correspond to evaluation points and columns to shape functions.

$$\left[N_i(P_j)\right]_{j=1,\dots,8}^{i=1,\dots,4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

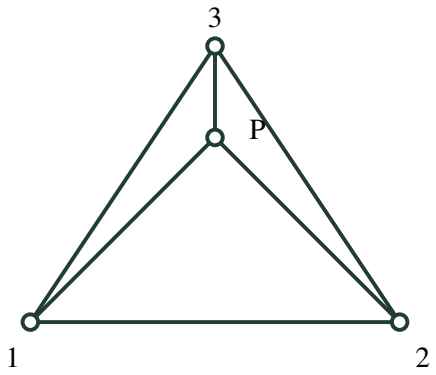
For example, at node 5, the factors $1 - 2x - y$ and $2x - 3y - 1$ both vanish, while $y = 0$ makes the two top functions vanish.

The full eight-function interpolation has partition of unity. The four corner functions are the subset requested here.

Question 3 · Linear triangle

Area coordinates of the triangle

Determine the three linear triangular shape functions at the point P = (1, 2) in the triangle with vertices (0, 0), (2, 0) and (1, 3).



Each shape function is the area of the subtriangle opposite the corresponding vertex, divided by the total area.

$$N_1(P) = \frac{A_{P23}}{A_{123}}, \qquad N_2(P) = \frac{A_{1P3}}{A_{123}}, \qquad N_3(P) = \frac{A_{12P}}{A_{123}}.$$

$$A_{123} = \tfrac{1}{2}(2)(3) = 3.$$

Question 3 · Linear triangle

Calculate the subareas and weights

For each subtriangle use rows (1, x, y) in counterclockwise order. Each determinant gives twice its area.

$$2A_{P23} = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 1 & 3 \end{vmatrix} = 6 - 3 - 2 = 1.$$

$$2A_{1P3} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{vmatrix} = 3 - 2 = 1, \qquad 2A_{12P} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 4.$$

$$(N_1, N_2, N_3) = \left(\frac{1}{6}, \frac{1}{6}, \frac{2}{3}\right).$$

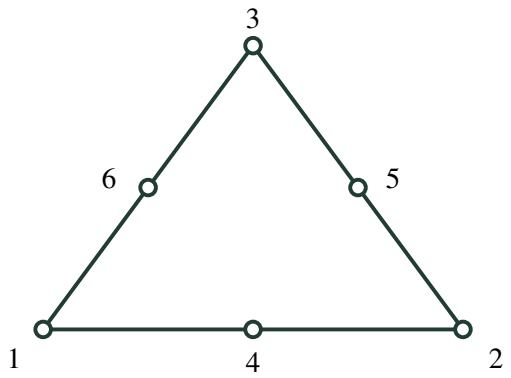
The weights sum to one. The interpolated coordinates recover the specified point.

$$x = 0(1/6) + 2(1/6) + 1(2/3) = 1, \qquad y = 0 + 0 + 3(2/3) = 2.$$

Question 4 · Quadratic triangle

Six-node triangular element

Determine the Jacobian and its determinant for the six-node linear-strain triangle (LST). Use the two independent coordinates r = L_2 and s = L_3, with $L_1 = 1 - r - s$.



Corner coordinates are (0, 0), (2, 0), (1, 2). The edge nodes are the physical midpoints of sides 1–2, 2–3 and 3–1, respectively.

$$P_4 = (1, 0), \quad P_5 = (3/2, 1), \quad P_6 = (1/2, 1).$$

The reference triangle is bounded by $r \geq 0$, $s \geq 0$ and $r + s \leq 1$.

Question 4 · Quadratic triangle

Write the interpolation functions

A corner function must be one at its vertex and zero at either neighbouring midpoint. The quadratic polynomial in its own area coordinate is therefore

$$N_i = L_i(2L_i - 1), \qquad i = 1, 2, 3.$$

Each edge function is a product of the two area coordinates for that edge. At the midpoint, each coordinate is one half, fixing the multiplier at four.

$$N_4 = 4L_1L_2, \qquad N_5 = 4L_2L_3, \qquad N_6 = 4L_3L_1.$$

$$x = \sum_{i=1}^{6} N_i x_i, \qquad y = \sum_{i=1}^{6} N_i y_i.$$

For example, N_4 equals one at the midpoint of side 1–2 and vanishes on the two sides where either L_1 or L_2 is zero.

Question 4 · Quadratic triangle

Simplify the map and differentiate

Substitute the six physical coordinates into the interpolation equations.

$$x = 2N_2 + N_3 + N_4 + \tfrac{3}{2}N_5 + \tfrac{1}{2}N_6, \qquad y = 2N_3 + N_5 + N_6.$$

$$\begin{aligned} x &= 2r(2r-1) + s(2s-1) + 4(1-r-s)r + 6rs + 2s(1-r-s) \\ &= 2r + s. \end{aligned}$$

$$y = 2s(2s-1) + 4rs + 4s(1-r-s) = 2s.$$

Use rows for physical coordinates and columns for reference coordinates.

$$\mathbf{J} = \frac{\partial(x,y)}{\partial(r,s)} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \qquad \det \mathbf{J} = 2(2) - 1(0) = 4.$$

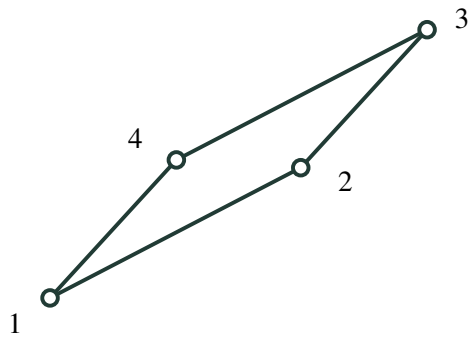
The area is 4 × 1/2 = 2 square units, consistent with the physical base and height.

Question 5 · Gauss quadrature

Quadrilateral and reference square

Determine the Jacobian of the four-node quadrilateral and evaluate the following integral by Gauss quadrature.

$$K = \int_{\Omega} \frac{\partial \psi_1}{\partial x} \frac{\partial \psi_4}{\partial x} \, \mathrm{d}x \, \mathrm{d}y.$$



Physical nodes 1–4 are (0, 0), (1, 1), (3/2, 2) and (1/2, 1). Assign the corresponding reference nodes in counterclockwise order.

$$(-1,-1), \quad (1,-1), \quad (1,1), \quad (-1,1).$$

Question 5 · Gauss quadrature

Map the square to the physical element

The four bilinear reference functions are

$$\psi_1 = \tfrac{1}{4}(1-\xi)(1-\eta), \qquad \psi_2 = \tfrac{1}{4}(1+\xi)(1-\eta),$$

$$\psi_3 = \tfrac{1}{4}(1+\xi)(1+\eta), \qquad \psi_4 = \tfrac{1}{4}(1-\xi)(1+\eta).$$

$$x = \psi_2 + \tfrac{3}{2}\psi_3 + \tfrac{1}{2}\psi_4, \qquad y = \psi_2 + 2\psi_3 + \psi_4.$$

Expanding the products and collecting terms gives

$$x = \frac{3}{4} + \frac{1}{2}\xi + \frac{1}{4}\eta, \qquad y = 1 + \frac{1}{2}\xi + \frac{1}{2}\eta.$$

At the centre, (ξ, η) = (0, 0) maps to (3/4, 1). Opposite sides are parallel, so the mixed term cancels in both coordinate maps.

Question 5 · Gauss quadrature

Jacobian and derivative transformation

Use the same convention as in Question 4, with physical coordinates listed down the rows. Its transpose is the matrix that transforms a physical gradient into a reference gradient.

$$\mathbf{J} = \frac{\partial(x,y)}{\partial(\xi,\eta)} = \begin{bmatrix} 1/2 & 1/4 \\ 1/2 & 1/2 \end{bmatrix}, \qquad \det \mathbf{J} = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}.$$

$$\begin{bmatrix} \psi_{,\xi} \\ \psi_{,\eta} \end{bmatrix} = \mathbf{J}^T \begin{bmatrix} \psi_{,x} \\ \psi_{,y} \end{bmatrix}, \qquad \mathbf{J}^{-T} = \begin{bmatrix} 4 & -4 \\ -2 & 4 \end{bmatrix}.$$

$$\psi_{,x} = 4\psi_{,\xi} - 4\psi_{,\eta}, \qquad \mathrm{d}x\,\mathrm{d}y = \frac{1}{8}\,\mathrm{d}\xi\,\mathrm{d}\eta.$$

Area check: the reference square has area 4, giving physical area 4/8 = 1/2.

Question 5 · Gauss quadrature

Differentiate the two shape functions

For the first corner function, differentiation in the reference coordinates gives

$$\psi_{1,\xi} = \frac{\eta - 1}{4}, \qquad \psi_{1,\eta} = \frac{\xi - 1}{4}.$$

$$\psi_{1,x} = 4\psi_{1,\xi} - 4\psi_{1,\eta} = \eta - \xi.$$

For the fourth corner function, the corresponding derivatives are

$$\psi_{4,\xi} = -\frac{1 + \eta}{4}, \qquad \psi_{4,\eta} = \frac{1 - \xi}{4}.$$

$$\psi_{4,x} = -(1 + \eta) - (1 - \xi) = \xi - \eta - 2.$$

The two derivatives are both taken in the global x direction, as required in the question.

Question 5 · Gauss quadrature

Choose the Gauss points

Include the area factor once in the reference integrand F.

$$F(\xi,\eta)=\frac{1}{8}(\eta-\xi)(\xi-\eta-2).$$

$$K=\int_{-1}^{1}\int_{-1}^{1}F(\xi,\eta)\,\mathrm{d}\xi\,\mathrm{d}\eta.$$

Each coordinate appears with degree at most two. A two-point Gauss rule integrates degree up to three in one coordinate. Use two points along each axis with unit weights.

$$g=\frac{1}{\sqrt{3}},\qquad \xi,\eta\in\{-g,+g\},\qquad w_{-}=w_{+}=1.$$

$$K=\sum_{i=1}^{2}\sum_{j=1}^{2}w_i w_j F(\xi_i,\eta_j).$$

Question 5 · Gauss quadrature

Evaluate and check the integral

At the four integration points, evaluate the integrand including the determinant. All products of weights equal one.

$$F(-g,-g)=0, \qquad F(g,-g)=-\frac{1}{6}+\frac{1}{2\sqrt{3}},$$

$$F(g,g)=0, \qquad F(-g,g)=-\frac{1}{6}-\frac{1}{2\sqrt{3}}.$$

$$K=0-\frac{1}{6}+\frac{1}{2\sqrt{3}}+0-\frac{1}{6}-\frac{1}{2\sqrt{3}}=-\frac{1}{3}.$$

As an independent check, expand the integrand. The linear terms and the mixed term integrate to zero on the symmetric square.

$$\begin{aligned} K &= -\frac{1}{8}\int_{-1}^{1}\int_{-1}^{1}(\xi^2+\eta^2)\,\mathrm{d}\xi\,\mathrm{d}\eta \\ &= -\frac{1}{8}\left(\frac{4}{3}+\frac{4}{3}\right)=-\frac{1}{3}. \end{aligned}$$